Mean correction in EEG decoding: decision geometry, temporal memory, and failure modes

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Abstract

Mean correction is a routine first step in cross-session EEG decoding, and whether it helps or hurts is usually assumed. For a fixed affine decoder only the projection of a displacement onto the centred weights can change a decision, and in a symmetric two-class model the balanced-optimal threshold is independent of the target class proportions: a composition-only correction must hurt, a drift correction can help, and the best weight is the drift projected onto the axis separating the two. Full-strength past-only correction gains 0.214 on SEED and 0.0705 on BCI-2a (exploratory; 0.034 on BCI-2b, pre-fixed) and loses 0.102 on sleep-EDF, negative in all 32 subjects. On the estimator actually used, the displacement splits exactly (error 2.7e-14) into composition, within-class and unmatched-class terms, anchored so that the parts do not depend on the feature origin; removing only the composition term costs 0.0715 and removing only the within-class term gains 0.0202. A controlled factorial on real feature statistics separates them: a composition change alone raises the cost from 0.0034 to 0.0097, a drift alone gains 0.0172 to 0.0601, and the sign follows whichever displacement is larger. Memory sets the scale: on SEED the sign changes between fifty and a hundred seconds of history, a quarter to a half of a film; on sleep-EDF the available history reduces the loss by 0.012 and never changes its sign. Filter and alignment act in opposite directions in a two-factor audit, and a grasp-and-lift negative case did not survive an audit of its cache order.

Keywords

cross-session EEG decoding; mean correction; origin equivariance; preprocessing audit; calibration; reproducibility

1. Introduction

A classifier trained on one session and applied to another usually loses accuracy, and subtracting the mean of the target session is the first thing most pipelines do about it. Whether that subtraction helps or hurts is usually assumed rather than checked. The paper asks when it helps and when it harms, and it answers with one mechanism in three parts: the decision geometry of the projection, the class composition and drift that produce the displacement, and the memory of the estimator relative to the class episodes.

Under the causal evaluation used here the sign is a property of the benchmark. Full-strength past-only correction gains 0.214 on SEED and 0.03-0.07 on the two motor-imagery datasets, and loses 0.102 on sleep-EDF with all 32 subjects negative. A second negative case, grasp-and-lift, did not survive an audit: its cache stored trials sorted by class, and in acquisition order the effect is +0.001.

Consistent implementation therefore does not guarantee a benefit: the direction of the displacement, not the correctness of the code, decides the sign. The three statements below each rest on a different kind of evidence.

Decision geometry. A shift of the feature origin changes decisions only through its projection onto the classifier weights, so a correction orthogonal to those weights cannot change a single prediction. This is an identity for affine decoders, and it bounds what any mean correction can do before any data are seen.

Correction strength. Given that a shift reaches the decision, how much to remove is a separate question. In a symmetric two-class model the balanced-optimal threshold does not depend on the target class proportions, so removing a composition-only displacement must hurt, while removing a genuine drift can help; the optimal strength is the projection of the drift onto the axis that separates the two.

Trial order. A running mean depends on the order in which trials arrive, and a cached table that is sorted by class rather than by time manufactures a spurious negative transfer. The audit that found this is part of the result, not an aside.

Two further results bound the claims: the one-sided form is not merely inconsistent but damaging in proportion to the origin offset, and no label-free rule tested here predicts the sign, so the sign analysis stays descriptive.

What this paper is not. It proposes no correction that is safe in general and does not identify the physiological cause of the composition change it localises. It offers a mechanism, its boundaries, and the evidence class of each claim.

Relation to prior work. Apicella et al. [4] found that a good normalisation often beats a domain-adaptation algorithm; our results say the amount of normalisation is the operative variable. Euclidean and Riemannian alignment [5-8] act on second-order structure and compose with the first-order correction studied here; SPDIM [23] shows that Riemannian alignment can hurt under label shift, the closest prior statement of the phenomenon, and differs from this work in object, analysis and scope. The label-shift estimator we use in S33 is Lipton et al. [24], imported rather than proposed. Test-time adaptation on EEG is independently reported to give inconsistent gains [25]; that benchmark measures accuracy, whereas we analyse the displacement that produces it.

Relation to our earlier work. An earlier manuscript by the same authors studied the consistency of the two centring conventions; this paper asks the different question of when the correction changes accuracy, and it replaces that analysis where the two overlap.

2. Decision geometry and the mechanism framework

2.1. Pipelines and the projection condition

Let A be the training session with features X_A and labels y_A and B the target session, and let the decoder be affine, s_k(x) = w_k'x + b_k with prediction argmax_k s_k(x) (Eq. 1). A mean correction subtracts an estimate of the target mean: the two-sided pipeline centres the target on its own mean and restores the training mean, the one-sided patch centres without restoring it, and the uncorrected pipeline does neither (Table 1).

$$s_k(x) = w_k'x + b_k , \quad k = 1 \dots K , \qquad (1)$$

Write W for the K x D matrix of weights, S for the matrix of score vectors and D = F(W (mu_B - mu_A)) for the displacement in decision space, with F the centring matrix that removes the mean score. Only the projection of the displacement onto the discriminative directions can change a decision; the full notation and the relation to standardisation are in S1.

Table 1. The pipelines compared in this paper.

name	fitted on	applied to	origin-equivariant
none	X_A	X_B	yes
zero	X_A	X_B − μB	no
mean	X_A	X_B − μB + μA	yes
dbl (refit)	X_A − μA	X_B − μB	yes
train-only	X_A − μA	X_B	no
z-score	(X_A − μA)/s_A	(X_B − μA)/s_A	yes

2.2. What equivariance does and does not give

Proposition 1. Under a common translation applied to both sessions with the classifier refitted, the uncorrected pipeline and the two-sided mean pipeline return identical predictions, and the one-sided pipeline does not. The two shifts cancel for the first two; the one-sided pipeline centres the target but never restores the training mean, so its scores move by -w_k'c (proof in S37). Table 1 records the same classification.

Proposition 2. A correction x -> x - gamma(X) is equivariant under a common translation if and only if gamma(X + c1) = gamma(X) + c for all c. The sample mean, the coordinate-wise median and a training-session reference plus a constant all satisfy it; the condition is a definition, not a characterisation of one estimator (proof in S37).

Proposition 3. A displacement d leaves every pairwise score difference unchanged for all inputs if and only if (w_j - w_k)'d = 0 for every pair of classes, that is F W d = 0 with F centring the score vector. The correction must lie in the null space of F W, not in the span of the weights (proof in S37).

For a finite target sample the condition is sufficient but not necessary: if no trial falls between the two decision boundaries the predictions can coincide even when the displacement is not in the null space of F W.

Proposition 4. Adding a constant to the scores of a class, or to all scores, leaves every within-class ranking and therefore every per-class AUC unchanged. The statement holds within one score function; it does not transfer between raw scores and pairwise margins, which need not order samples alike (proof in S37).

2.3. What a first-order correction can represent

What the displacement contains. Write the source class proportions pi_A and means m_A, the target proportions pi_B and means m_B. The correction removes mu_B - mu_A, which splits into a composition term sum_c (pi_B,c - pi_A,c) m_A,c and a within-class term sum_c pi_B,c (m_B,c - m_A,c). The first is present even when no class-conditional distribution changes, the second vanishes when it does not, and a correction cannot separate them without knowing the labels.

The consequences are quantitative in a symmetric two-class model. With X | Y = +1 ~ N(a, s^2) and X | Y = -1 ~ N(-a, s^2), balanced accuracy at threshold t is BA(t) = (Phi((a - t)/s) + Phi((a + t)/s))/2, so BA"(0) = -a phi(a/s)/s^3 and a displacement d costs BA(0) - BA(d) = a d^2 phi(a/s)/(2 s^3) to second order. The balanced-optimal threshold is t = 0 for every target mixture, which is why a composition-only correction cannot help in this model. With a drift b and a composition displacement c = (2 pi_B - 1) a the residual after correcting at strength lambda is (1 - lambda) b - lambda c, and the optimal strength is clip_[0,1](b/(b + c)), undefined when b + c = 0. A Monte Carlo over 27 parameter cells reproduces the second-order coefficient to within 0.9 per cent at d = 0.1 (S2).

What stays exact in the multi-class case is a per-sample stability certificate. Write s_k(x) for the uncorrected score and s'_k(x) = s_k(x) - d_k for the corrected one. If k* is the class predicted before the correction, the prediction is unchanged exactly when s_k*(x) - s_j(x) > d_k* - d_j for every j != k*, with a non-strict inequality when scores tie; both scores on the left are uncorrected, because the displacement has been moved to the right-hand side. Equivalently, in corrected scores, s'_k*(x) - s'_j(x) > 0. The certificate is an identity for a fixed affine decoder and predicts stability, not accuracy.

3. Data and protocol

Datasets. Five public datasets span tasks, feature families and session structures; Table 2 lists their sizes, class counts and origin distances.

BCI Competition IV 2a [13]: four motor-imagery classes, 22 channels, 250 Hz, nine subjects, two sessions on different days.

BCI Competition IV 2b [13]: two classes, three channels, nine subjects, screening and feedback sessions on different days.

SEED [14,15]: three emotion classes, 62 channels, 15 subjects, three sessions on different days, with the released 1 s differential-entropy features in 15 film-defined blocks.

NER2015: sixteen subjects, five sessions, two classes, correct versus error feedback.

Grasp-and-lift: twelve subjects, eight series in one session, 28-34 grasp trials per series.

SEED-IV [19]: 15 subjects, four classes, three sessions; examined during an earlier implementation audit and reported in the supplement.

Table 2. The datasets. The last column is z_T = ‖μ_A‖/‖sd_A‖, the norm-based origin distance, median over cells.

dataset	classes	subjects	sessions	representations	cells	median z_T
BCI-IV-2a	4	9	18	4	36	0.50
BCI-IV-2b	2	9	27	4	108	16.1
SEED	3	15	45	1	45	34.2
NER2015	2	16	80	1	320	0.20
grasp-and-lift	2	12	12	1	336	0.25
SEED-IV	4	15	45	1	90	56.3

A cell is one dataset-representation-subject-ordered-pair combination: 845 development cells from 61 subjects and 182 sessions, 737 of them session pairs. The online protocol scores trials 16 onwards, the first block of 15 being warm-up, and the correction for block k uses only blocks 1 to k-1.

Representations. euc is a per-trial log variance, csp uses filters fitted on the training session only, tan maps the session covariance to tangent space, and euc_ea divides by the session covariance. All but tan and euc_ea are source-only, which is what a strictly causal pipeline can use.

Classifiers. Linear discriminant analysis with Ledoit-Wolf shrinkage [11] throughout, with logistic regression and a Riemannian classifier as sensitivity checks in the supplement.

Two adapted baselines. Adaptive-window estimation follows ADWIN [21], and a predictive rule weights each block by its leave-one-block-out reliability. Both are reported with the same warm-up and the same scored trials as the estimators they are compared with.

Reference. The within-session reference is stratified cross-validation inside the target session using its own labels; it is an upper bound, not a deployable method.

Statistics. Cells are aggregated per subject first, then tested across subjects with a two-sided Wilcoxon signed-rank test, a subject-level bootstrap of the mean (10,000 resamples) and Cohen dz. Cell-level correlations are descriptive, and no analysis treats the 845 cells as independent.

Which tests are primary. Five tests were pre-specified as one family with Holm control; after correction the p-values are 3.05e-4 (SEED), 0.016 (BCI-2a), 0.016 (BCI-2b), 0.104 (grasp-and-lift) and 0.320 (NER2015), so the gain survives on three datasets and the family is not five confirmations of one claim. Everything outside it is exploratory.

Preprocessing. The 2a/2b features use a fourth-order Butterworth 8-30 Hz band-pass; 2a is epoched 2.0-6.0 s after the trial onset and 2b 0.5-3.5 s after the cue, a protocol difference between the datasets rather than a common setting. LDA uses solver=lsqr and shrinkage=auto.

4. Results

Overview of the findings. Within a single recording, full-strength past-only mean correction can be harmful: on sleep-EDF - a task the analysis never touched - it lowers balanced accuracy by 0.102 over 4,983 cells, negative in all 32 subjects and in all 45 recordings (Section 4.6). It can also help: SEED gains 0.214 and the two motor-imagery datasets about 0.03. A third within-recording benchmark, grasp-and-lift, was the paper's original negative case; after its cache was rebuilt in acquisition order the effect disappeared (+0.001, Section 4.5), and the class-sorted storage that produced the original -0.108 is reported as a methodological result in its own right. The cause of the sleep-EDF loss is not identified: class composition covaries with time, and a cell-level partial correlation leaves no residual association (the Discussion). The external cross-day cohort is positive under the executed window but not under the development pipeline (+0.010, $p = 0.10$), so it establishes no gain (Section 4.8).

Evidence tiers. The pre-specified family is the five per-dataset Wilcoxon comparisons of mean against none, Holm-corrected (Section 3); the gain survives on three of five. It is a family of tests, not a confirmation of a single causal mechanism, for the reasons just given. The strength sweep, the estimator-by-strength surface, the trend association, the block-type diagnostics, the two-factor protocol audit and the two independent corpora are all reported outside that family, each with its own evidence class, and none of them is used to strengthen the family's conclusion.

4.1. Two descriptions of the same pipeline, before the question of strength

Two descriptions of the same pipeline. Fitting on $X_A - \mu_A$ and applying to $X_B - \mu_B$, versus fitting on X_A and applying to $X_B - \mu_B + \mu_A$, gave identical predictions in every cell (Table 3): the two-sided pipeline is unchanged when both sessions are translated. Standardisation with training statistics is a different operation - in the space $(X - \mu_A)/s_A$ the source mean is zero and the target mean is not - so it is a one-sided correction in disguise.

Table 3. Transductive correction: balanced accuracy per dataset. dbl coincides with mean in every cell. The reference is the within-session cross-validation of Section 3.

dataset	n	none	zero	mean	dbl	train-only	reference
BCI-2a (K=4)	36	0.5656	0.5254	0.6119	0.6119	0.5141	0.5355
BCI-2b (K=2)	108	0.7046	0.5831	0.7375	0.7375	0.5722	0.7571
SEED (K=3)	45	0.4591	0.3377	0.7094	0.7094	0.3456	0.7738
NER2015 (K=2)	320	0.5886	0.5865	0.5909	0.5909	0.5845	0.6167
grasp-and-lift (K=2)	336	0.7770	0.7484	0.7815	0.7815	0.7521	0.8350
all development	845	0.6705	0.6346	0.6926	0.6926	0.6338	0.7264

The gain of mean over none is +0.0463 on 2a, +0.0329 on 2b and +0.2503 on SEED, and small where the features carry no free origin (Fig. 1).

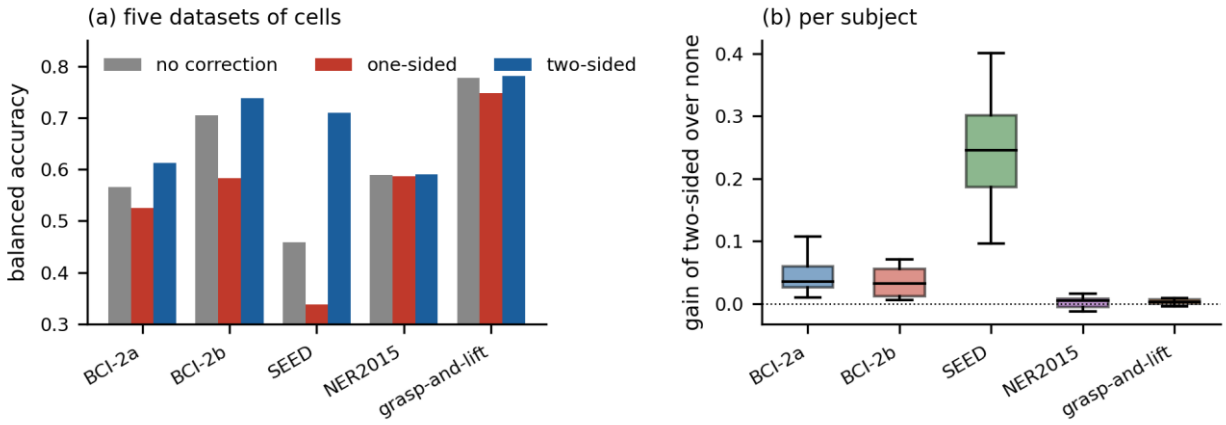


Fig. 1. (a) Balanced accuracy of the three pipelines by dataset; (b) the gain of the two-sided over the uncorrected pipeline, one point per subject, with the box showing the median and quartiles.

4.2. Causal estimation: an implementation check, a comparison, and the gates that failed

Three things are separated here: whether a pipeline is invariant to the origin at all, which causal estimator is best on a common test set, and whether a rule can decide in advance whether to correct. Adding a common constant to every feature in both sessions and sweeping its size over ten settings on all 845 cells shows that the equivariant pipelines are unchanged while the one-sided pipeline moves with the origin (Fig. 2).

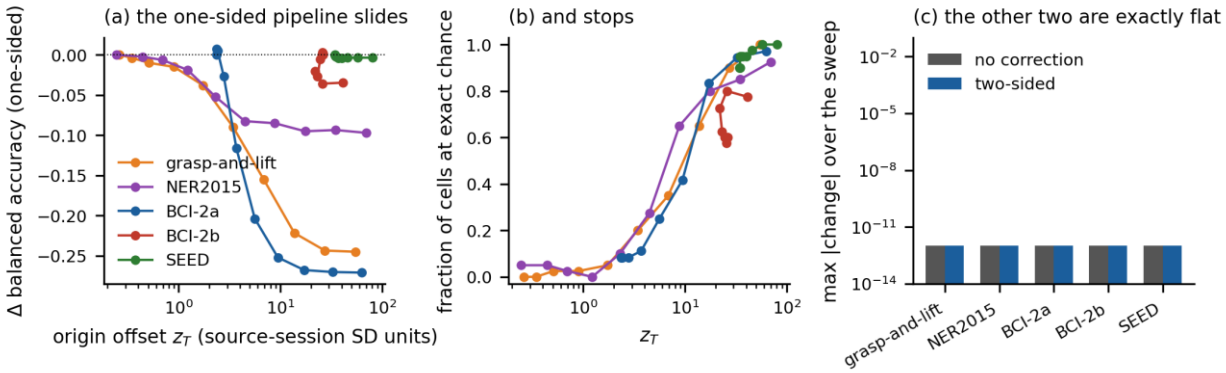


Fig. 2. (a) Change in balanced accuracy of the one-sided pipeline as a function of z_T, one curve per dataset; (b) the fraction of cells that land on exactly 1/K; (c) the largest change of the two equivariant pipelines over the whole sweep, on a logarithmic axis whose floor is 10^{-16}, so that a zero is drawn at the floor rather than as a measured value. The equivariant pipelines change by 0.0e+00 in every cell at every setting; the one-sided pipeline decays toward 1/K without reaching it uniformly within the sweep.

4.2.1 Estimators on a common test set.

The comparison that follows uses source-only representations; the target-informed ones are in S35, together with the two comparison tables.

The mean has to be estimated online, and the estimator is a design choice: the two-sided form corrects by the past estimate and restores mu_A (Eq. 7), the one-sided patch does not (Eq. 8), and the two differ by exactly the quantity at issue.

$$\tilde{x}_t = x_t - \hat{\mu}_{t-1} + \mu_A \,, \qquad (7)$$

with $\hat{\mu}_{t-1}$ an estimate from trials 1 … t−1; as $\hat{\mu} \to \mu_B$ it converges to `mean`. The *one-sided patch* is the same without the training mean,

$$\tilde{x}_t = x_t - \hat{\mu}_{t-1} \,, \qquad (8)$$

Four estimators are compared under one warm-up: a cumulative mean, a window of 50 trials, an exponential average with alpha = 0.05, and a shrunk cumulative mean. The full ranking is in S34.

A block-wise estimate weighted by its reliability corrects by the mean of the completed blocks, weighted by their share of signal variance (Eq. 9); because every d_b is translation invariant, so is the weight.

$$\lambda = \|\bar{d}\|^2 / (\|\bar{d}\|^2 + K\, V_d / B) \,, \qquad s_{corrected} = s - \lambda\, \bar{d} \,, \qquad (9)$$

where V_d is the mean over completed blocks of the within-block variance of the per-trial decision-space displacement d_i (averaged over its K components), B is the number of completed blocks, and $\bar{d}$ is their mean; the factor K restores the scale that the projection onto K centred scores removes.

Once the window is fixed the SEED comparison is unambiguous: the variance-ratio rule keeps +0.211 of the +0.250 that the transductive correction achieves there, gains about 0.03 on the two motor-imagery datasets and is flat on NER2015.

4.3. Synthetic analysis of correction failure regimes

A controlled generator asks why. The same estimator wins under a constant offset and loses under a drift, and the transition depends on the class proportions (Fig. 3, Table 4); the controlled factorial of Section 4.7 extends this to the real feature statistics, so this section is retained as the origin of the regime map.

Table 4. Controlled regimes: absolute balanced accuracy per method, mean over 30 paired seeds (the same seeds for every method). a is the constant offset and v the total drift, both in units of the mean distance between class means; the last three rows change the class or block composition instead. Class separation is calibrated so that the uncorrected accuracy is 0.70 in the no-perturbation condition; under a perturbation the uncorrected accuracy differs by construction, so the gain of a method is its value minus the none column of the same row. The generator, its seeds and its raw output are in the repository.

regime	none	cumulative	Eq. (9)	predictive	ADWIN
a = 0, v = 0	0.701	0.699	0.701	0.701	0.698
a = 1, v = 0 (offset only)	0.500	0.699	0.699	0.693	0.698
a = 0, v = 1 (drift only)	0.619	0.675	0.667	0.672	0.682
a = 0, v = 2 (strong drift)	0.519	0.617	0.602	0.611	**0.663**
a = 1, v = 2 (offset + drift)	0.379	0.617	0.609	0.607	**0.663**
steps every 8 blocks	0.342	0.440	0.432	0.430	**0.607**
class priors only (0.4)	0.704	0.691	0.697	0.695	0.690
block priors, no offset	0.706	0.693	0.698	0.698	0.688

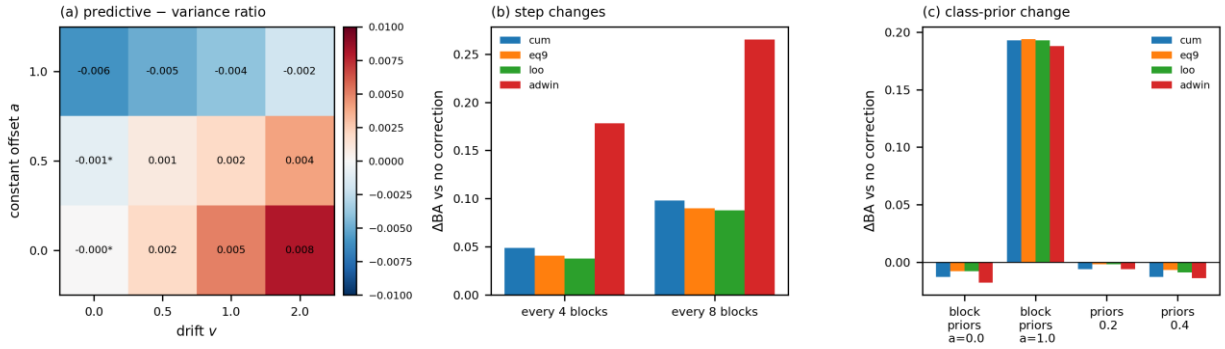


Fig. 3. The failure-regime map. (a) Predictive minus variance-ratio accuracy across the constant-offset and drift grid; blue means the variance-ratio rule is better, red the predictive rule, an asterisk marks cells whose paired 95 per cent interval crosses zero. (b) Step changes: change from no correction for four rules. (c) Class-prior change, without a constant offset (two left-hand bars) and with an offset of one class-distance (third bar); the fourth bar is the prior-shift condition without block structure. The axis therefore mixes two conditions and the bars are not a dose-response.

4.4. Correction strength and estimator memory: dataset-dependent effects

Correction strength. Every method above changes the same displacement, so the strength isolates how much of it is removed. On SEED the estimator family decides the sign: the long-memory rules gain 0.21 at full strength while the windowed and exponential rules lose, the two BCI datasets are insensitive at this resolution and NER2015 is flat. The subject-level full-strength contrast is SEED +0.214 [+0.167, +0.260], BCI-2b +0.032 [+0.017, +0.048], BCI-2a +0.029 [+0.012, +0.048] and NER2015 -0.004 [-0.010, +0.002]; the film-disjoint check leaves +0.189, so the SEED gain is not memorised film identity. Grid optima were chosen with target labels, so the figure reports what the benchmarks do, not what a deployment could achieve.

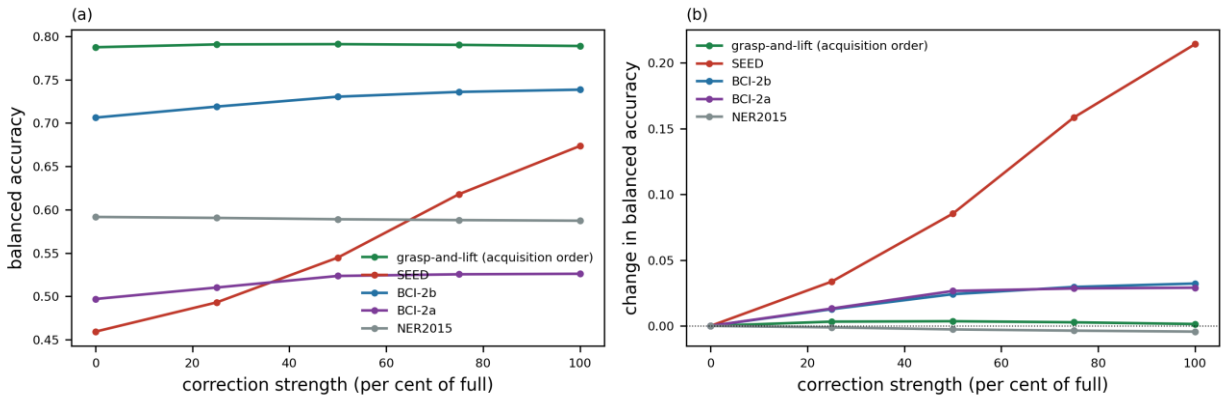


Fig. 4. Balanced accuracy as a function of the correction strength λ, cells aggregated within subject and bootstrapped over subjects (6,000 resamples, band = 95 per cent). (a) absolute; (b) change from no correction. The grasp-and-lift curve uses the acquisition-order cache of Section 4.5; the class-sorted cache is the source of the loss that curve no longer shows.

Table 5. Causal online correction: the correction-strength response, subject level. D = BA(λ = 1) − BA(λ = 0), averaged within subject and bootstrapped over subjects.

dataset	λ=0	0.10	0.25	0.50	0.75	λ=1	D [95% CI]	subjects with D < 0
grasp-and-lift (acquisition order)	0.787	0.789	0.791	0.791	0.790	0.789	+0.001 [-0.005, +0.008]	6 / 12
SEED	0.459	0.473	0.493	0.545	0.618	0.674	+0.214 [+0.167, +0.260]	0 / 15
BCI-2b	0.706	0.713	0.719	0.730	0.736	0.739	+0.032 [+0.017, +0.048]	1 / 9
BCI-2a	0.497	0.502	0.510	0.524	0.526	0.526	+0.029 [+0.012, +0.048]	1 / 9
NER2015	0.592	0.592	0.591	0.589	0.588	0.587	−0.004 [−0.010, +0.002]	10 / 16

4.5. Trial order in a cached feature table, and a negative transfer that came from it

The comparison in Section 4.5 used a cached feature table in which the samples of each series were stored sorted by class: every one of the 96 grasp-and-lift series held 28 or 34 class-0 samples followed by the same number of class-1 samples, with the switch at exactly half the file. The recordings do not have that structure. Rebuilding the same features in acquisition order changes the result completely: the published cache gives -0.108 with all twelve subjects negative, the same features stored class-sorted give -0.090, again in twelve of twelve, and in acquisition order +0.001 with six subjects improving and six worsening (Table 6). The remaining grasp-and-lift numbers

computed on the sorted cache are withdrawn; the acquisition-order values are in Table 6. A control removes the last difference between the two orderings: because the warm-up block is excluded, the two orders do not score the same trials. Restricting both to the trials scored in either order leaves the effect intact - -0.001 in acquisition order against -0.114 in the sorted order on the identical trial set, the sorted order lower in 292 of 336 cells.

Table 6. The same estimator on three orderings of the grasp-and-lift cache: the published cache, the rebuilt features stored class-sorted, and the rebuilt features in acquisition order. Interval: subject-level bootstrap, 10,000 resamples.

cache	BA at lambda = 0	BA at lambda = 1	delta [95 per cent interval]	subjects negative
published cache (class-sorted)	0.7727	0.6652	-0.108 [-0.132, -0.083]	12 of 12
same features, class-sorted	0.7925	0.7025	-0.090 [-0.105, -0.075]	12 of 12
same features, acquisition order	0.7874	0.7889	+0.001 [-0.005, +0.008]	6 of 12

The mechanism has nothing to do with EEG non-stationarity. In the sorted order the target sequence changes class once, at its midpoint, so a running mean over blocks of 15 carries a large class-prior step in the second half of every cell and the correction shifts the scores away from the minority class. In acquisition order every block of 15 covers about seven trials and contains both classes, so the running mean is close to zero and the correction is nearly inactive. The block-composition analysis of Section 4.5 describes this structure exactly; its subject is the cache, not the recording.

What the block mean contains. The decomposition was computed on the class-sorted cache and is kept because it explains the artefact of Table 6. Writing mu_b = sum_k pi_bk mu_bk splits the displacement exactly into a class-prior term sum_k (pi_bk - pi_sk) mu_sk and a class-conditional term sum_k pi_bk (mu_bk - mu_sk); the reconstruction is exact to 1e-16, and the prior term accounts for a median +1.27 of the movement against -0.27. In the sorted order that prior term is the step at the midpoint of every series, which is why full-strength correction moves the second half of each cell so far: the composition analysis and the artefact are the same object seen from two sides.

4.6. Mechanism I: which part of the displacement does the damage

The mechanism analysis has to use the displacement the predictor actually subtracts, and it has to be independent of where the feature origin happens to sit. For scored block k the estimator removes the mean of the completed blocks before it, d_k = mu_hat(H_k) - mu_A with H_k the blocks 1 to k-1. Writing pi_hat_c for the block-mean-weighted class proportion over H_k and m_hat_c for the matching class mean, sum_c pi_hat_c m_hat_c = mu_hat(H_k) exactly. Anchoring every term on the source mean gives three terms that sum to d_k and are each invariant under adding a constant to every feature: a composition term sum_{c in A} (pi_hat_c - pi_A,c)(m_A,c - mu_A), a within-class term sum_{c in A} pi_hat_c (m_hat_c - m_A,c), and a residual sum_{c not in A} pi_hat_c (m_hat_c - mu_A) for classes absent from the source. Over 4,983 cells and 14,949 scored blocks the three terms reproduce d_k to within 2.7e-14. The anchoring is not cosmetic: with an unmatched class of mass u in the history, the unanchored composition term moves by -u a under a translation a and the residual by +u a, so with no fixed origin only their sum is defined. A test with an unmatched class in the history confirms that the anchored terms move by less than 1e-9 under a translation of four units while the unanchored ones move by more than 1e-3.

Correcting with one component at a time, on the same classifier, the same scored epochs and the same warm-up, gives the following subject-level means. The full correction costs 0.1023 of balanced accuracy, interval [-0.1141, -0.0898], negative in all 32 subjects. Removing only the composition term costs 0.0715, interval [-0.0792, -0.0635], negative in all 32. Removing only the within-class term gains 0.0202, interval [+0.0151, +0.0253], positive in 29

of 32. Removing the composition and within-class terms together costs 0.0759, and the unmatched residual alone costs 0.0148. About half of the scored blocks have no unmatched class in their history at all. The stage-wise recalls show where the loss falls: wake rises from 0.535 to 0.545 while N1 falls from 0.397 to 0.253, N2 from 0.752 to 0.565, N3 from 0.679 to 0.623 and REM from 0.728 to 0.526, so the correction buys a little wake at the cost of the sleep stages. It changes the decision on 39.3 per cent of scored epochs. Cohen dz is -2.84 for the full correction, -3.09 for the composition term alone, +1.36 for the within-class term, -2.58 for the two together and -1.41 for the residual.

An earlier version of this analysis decomposed the whole-target displacement rather than the estimator's own d_k, and a model-dependence check applied the same shift to a margin-based decoder. Both are kept in S31 and S24 as sensitivity checks; neither is the evidence used in this section.

Two cautions are structural rather than rhetorical. First, the components are not additive in balanced accuracy: the signed sum of the three component effects is -0.0715 + 0.0202 - 0.0148 = -0.0661 against the -0.1023 of the full correction, so the components do not compose additively; the 0.0362 gap mixes the interaction between the terms with the non-linearity of balanced accuracy as a functional of the scores, and the components are therefore reported separately and never added. Second, the unanchored split used in the previous version of this analysis gives 0.1257 for the composition term, nearly twice the anchored value, because the unmatched residual and the composition term trade an origin-dependent amount; those numbers are kept in the supplement as a diagnostic and are not used. Third, the history labels enter the decomposition only and never the predictor: the analysis locates the composition term as the part of the displacement that does the damage, and says nothing about why the stage proportions change across a night.

4.7. A controlled intervention separating composition from drift, and the role of memory

The decomposition is observational, so the two factors were also manipulated directly. Class-conditional means and a pooled covariance were estimated from the real sleep-EDF features of twelve subjects, and semi-synthetic series were drawn from them with a controlled class proportion and a controlled additive drift, keeping the decoder, block length, warm-up and correction rule unchanged. The design is 2 x 3: two levels of the composition factor (unchanged, or +0.25 mass moved to one class and renormalised) and three levels of the drift factor (none, 0.5 and 1.0 standard deviations of a random-direction offset normalised by sqrt(trace(cov)/d)/sqrt(d)), with twenty seeds per cell and 1,440 runs.

With no change and no drift the correction still costs accuracy: -0.0034 [-0.0046, -0.0020], so subtracting a noisy estimate has a price of its own. A composition change alone makes it worse, -0.0097 [-0.0112, -0.0082], a main effect of -0.0064 ($p = 0.001$). A drift alone produces a gain, +0.0172 [+0.0127, +0.0224] at half a standard deviation and +0.0601 [+0.0487, +0.0725] at one, positive in all twelve subjects at both magnitudes, a main effect of +0.0634 ($p = 0.0005$). The interaction is small and positive, +0.0020 ($p = 0.027$). Pooling runs, the sign is predicted by which displacement is larger in score space: where the drift exceeds the composition term the mean effect is +0.0398, where it does not, -0.0057. With both present the gain is +0.0121 at half a standard deviation and +0.0557 at one, so the composition change trims the drift benefit without removing it. The inference unit is the subject: seeds and repetitions are averaged within a subject, so the twelve subjects, not the 240 runs, are the sample.

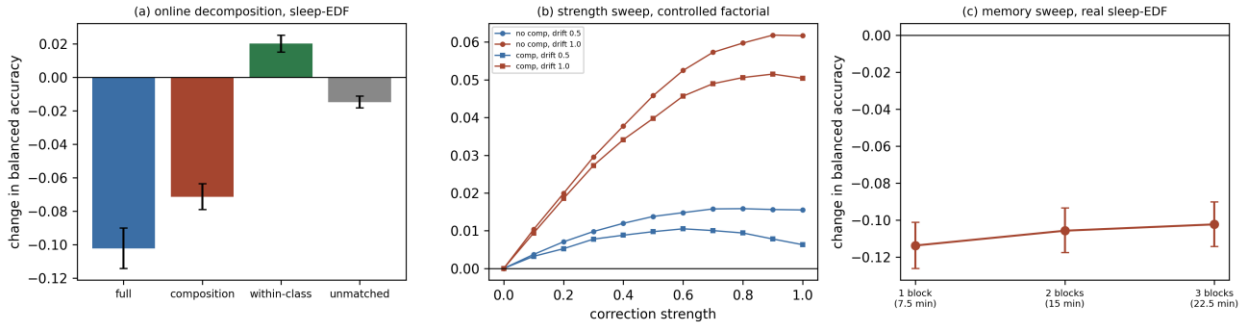
This is the ordering the symmetric Gaussian model predicts, and the strength grid tests the second half of the theory. With no drift and no composition change every increase in strength loses accuracy, from 0 to -0.0053 at full strength; with a composition change and no drift it loses more, -0.0119. With a drift the curve turns over: on a grid of eleven strengths the measured optimum is 0.8 [0.7, 1.0] for half a standard deviation without a composition change, 0.9 [0.9, 1.0] for one standard deviation, 0.6 [0.5, 0.8] with the composition change at half a standard deviation, and 0.9 [0.8, 0.9] with it at one. The closed-form optimum b/(b+c) is 1.00, 1.00, 0.60 and 0.75 for the same cells. With a composition term present both intervals exclude 1.0; without one both include it, so the measured optimum falls below full strength when a composition term is present, and that is a tendency with a clear boundary rather than a rule.

Memory is the axis behind both patterns. On SEED, where a segment is one second, the effect moves from -0.1129 [-0.1453, -0.0804] at five seconds of memory to +0.2025 [+0.1624, +0.2435] at four hundred, positive in all fifteen subjects, and the sign changes between fifty and a hundred seconds - a quarter to a half of the 226-

second film. That supports a memory account; it does not support the stronger reading that the memory must span a whole class episode, which is not tested here.

Sleep-EDF sits at the other end of the scale: stage runs have a median of 1.5 minutes and the wake stage occupies 68 per cent of a night, while the memory is one to three blocks of 7.5 minutes. Limiting the correction to one, two or three blocks gives -0.1138 [-0.1262, -0.1012], -0.1057 [-0.1178, -0.0931] and -0.1023 [-0.1144, -0.0902], all with 32 of 32 subjects negative: longer memory reduces the loss by 0.012 and never changes its sign, because a 60-epoch series holds only three blocks. Section 4.7 shows what a longer memory would do; this corpus cannot reach that regime.

Fig. 5. The three mechanism results. (a) Online decomposition of the past-only displacement on sleep-EDF: the full correction, the composition term alone, the within-class term alone and the unmatched residual, subject-level means with bootstrap intervals. (b) Correction strength against the change in balanced accuracy in the controlled factorial, for the four combinations of composition change and drift magnitude; the optimum falls below full strength when the composition term is present. (c) Memory sweep on the real sleep-EDF corpus: longer memory reduces the loss by 0.012 and never changes its sign.



4.8. Independent validation: a second recording structure and a cross-day cohort

Sleep-EDF, within recording. Sleep-EDF took no part in development: one recording is cut into consecutive 30-minute series, source earlier and target later, scoring five sleep stages from two channels with blocks of 15 epochs. Forty-five recordings from 32 subjects gave 4,983 cells; the full-strength correction lowered balanced accuracy by 0.102 ([-0.115, -0.090]), negative in all 32 subjects (p = 4.7e-10) and in all 45 recordings (p = 5.7e-14). The loss survives strict causal filtering (-0.102 [-0.114, -0.089]), survives restriction to cells with identical class sets (-0.061, 29 of 32 negative), grows when single-stage targets are included (-0.152), and is not specific to the completed-block estimator: the cumulative, windowed, exponential and variance-ratio means lose 0.107, 0.108, 0.052 and 0.082, each negative in 32 of 32 (S24).

A second acquisition protocol. The same corpus contains a telemetry sub-study recorded five years later with different hardware (22 subjects, 44 recordings), with a protocol fixed before the run. It gives 3,945 cells and a difference of -0.048 ([-0.059, -0.037]), negative in 21 of 22 subjects (p = 1.1e-5) and in 30 of 44 recordings. The loss is about half that of the cassette protocol, so its magnitude depends on the recording structure while its sign does not; the telemetry recordings are the same family as the discovery data, not a new cohort.

A cross-day cohort where the gain is not established. On Lee2019-MI (54 subjects, day 1 as source and day 2 as target, fifteen unscored warm-up trials) the paired difference is +0.032 [+0.016, +0.048] under the reference window and +0.010 (p = 0.098) under the development pipeline. The direction is stable and the magnitude is not: the cohort does not establish a gain across days, and the executed number is reported as one variant of the sensitivity analysis.

4.9. Filtering and alignment: a two-factor protocol audit

Protocol effects, separated. The archived pipeline filters with a zero-phase filter, which reads future samples, and builds the alignment matrix from both sessions. A 2x2 design on BCI-IV-2a and 2b crosses filtering (causal or zero-phase) with the alignment reference (source only or both sessions), at fixed strength with identical scored trials.

On BCI-2a (nine subjects, euc_ea) the correction gains 0.0705, exploratory because the representation was chosen after a first run; causal filtering costs 0.011 ($p = 0.13$) and the both-session alignment gains 0.008 ($p = 0.65$), in opposite directions, so a near-zero total is consistent with two changes that cancel.

On BCI-2b the endpoint was fixed in advance: csp under causal filtering with source-only alignment gains 0.0341 [+0.0104, +0.0585], positive in eight of nine subjects, Wilcoxon $p = 0.039$ uncorrected. It is prospective on a dataset already used in development, not an independent confirmation. The filter factor on 2b is -0.0058 ($p = 0.25$), and the stability certificate of Section 2.3 predicted the flip pattern on all nine subjects (793 scored trials, flip rates 0.03-0.36). Recomputing the endpoint at block lengths 10 and 20 gives +0.0336 and +0.0340 with intervals that still exclude zero but $p = 0.055$ in both cases: the size is insensitive to the block length while the rank test sits on the boundary, and the endpoint is reported as a stable effect with a borderline test.

Table 7. Every headline result with its evidence class and the limits of what it establishes.

Result	Estimate	Evidence class	What it does not establish
SEED correction gain	+0.214 [+0.167, +0.260] archived; +0.236 [+0.204, +0.271] recomputed, 15/15	past-only on released DE features	not an end-to-end causal test; preprocessing not controlled
BCI-2a euc_ea gain	+0.071 [+0.041, +0.104]	EXPLORATORY	representation chosen after a first run
BCI-2b csp gain	+0.0341 [+0.0104, +0.0585], p = 0.039 uncorrected	PROSPECTIVE-IN-EXPLORED-DATA	dataset used in development; not independent
Sleep-EDF cassette loss	-0.102 [-0.115, -0.090], 32/32	CONFIRMATORY within the corpus	one corpus; within-recording transfer only
Sleep-EDF telemetry loss	-0.048, 21/22	replication, same corpus	not an independent dataset
Lee2019-MI cross-day	+0.010, p = 0.098	development pipeline	gain not established
P300 speller	-0.0005 [-0.0024, +0.0012], 7/16	EXPLORATORY-INDEPENDENT	interval inside a 0.02 margin, but decoder at chance
BCI-challenge ERRp	-0.0044 [-0.0166, +0.0075], p = 0.45	EXPLORATORY-INDEPENDENT	decoder near chance; sign not tested
Grasp-and-lift artefact	+0.001 acquisition order vs -0.108 sorted	CONFIRMATORY	about the cache, not about EEG

Table 8. Mechanism evidence matrix. Each mechanism with what the theory gives, what the real corpora show, what the controlled experiment adds, and what none of them establishes.

Mechanism	Theory	Real EEG evidence	Controlled evidence	Cannot show
Projection onto the discriminative direction	identity for affine decoders	the same correction gains on SEED and loses on sleep-EDF	not applicable: it is an identity	which sign a new benchmark will show
Class composition	composition-only correction must hurt in the symmetric two-class model	composition term costs 0.0715, within term gains 0.0202	composition alone raises the cost from 0.0034 to 0.0097	the physiological origin of the change
Genuine drift	optimal strength clip(b/(b+c))	conditional gains on SEED, close to zero on P300 and ERRp	drift alone gains 0.0172 to 0.0601; measured optimum 0.6-0.9	that real drift is a common offset
Estimator memory	the past mean matches the present only if the displacement is stationary over the window	SEED crosses between 50 and 100 s; sleep-EDF -0.1138 to -0.1023 with no crossing	memory sweep on real data, one to three blocks; the strength grid varies strength, not memory	a universal optimal memory
Trial order	a sorted cache destroys the online history	acquisition order restores +0.001 against -0.108	not applicable: the artefact is in the cache	a brain mechanism: it is a protocol artefact

5. Discussion

The three mechanisms are one argument. A correction acts only through the projection of its displacement onto the discriminative direction, so what matters is not how large the subtracted mean is but how much of it lies along the axis that separates the classes. Class composition and a genuine drift both move the mean, in ways that can oppose one another: composition contributes a displacement the balanced-optimal threshold does not want, drift contributes one it does. That is why the same operation helps on SEED and hurts on sleep-EDF without either corpus being anomalous.

What the evidence supports. The decomposition is exact for the estimator actually used (2.7e-14), it is anchored on the source mean so that its parts do not depend on the feature origin, and it localises the damage in the composition term (-0.0715) while the within-class term is mildly beneficial (+0.0202). The controlled factorial reproduces the direction of both effects and places the measured optimum strength below one when a composition term is present.

What remains a hypothesis. Why the stage proportions drift across a night is not identified; the decomposition uses target labels and is an oracle diagnostic, not a deployable rule. The factorial establishes that the mechanism is sufficient, not that the real corpus contains a drift of the assumed size or direction. Memory is one axis: the two corpora sit at different points of the same curve, but task, class count, feature set and split differ as well.

Limits. Prevalence was audited on three released components, not across the field. The source-only statistic is not a screening rule: five datasets, inconsistent leave-one-out sensitivity. The displacement model is quantitative for mean-calibrated linear classifiers, directional for the Gaussian parameterisation and not applicable to margin-based classifiers. Both protocols estimate the target mean from the whole target session, which an inductive deployment cannot do; the causal experiment maps that limit rather than removing it.

Statistical scope and reproducibility. The sample is hierarchical - 845 cells from 61 subjects and 737 session pairs - and every test is aggregated at the subject level. Five development datasets give little power to any dataset-level statement, and the two external corpora are reported with intervals and an equivalence margin rather than as confirmations. Every numerical summary was regenerated from the cell-level tables and five executable tests accompany the release: no future information reaches a scored block, the decomposition reconstructs the displacement, every condition is scored on the same epochs, the variance-ratio weight stays in [0, 1], and the

anchored components are translation invariant. The chain from raw recordings to the cache was not rerun, and the SEED archived figure could not be regenerated exactly (+0.214 archived against +0.236 recomputed).

6. Conclusion

A mean correction is neither beneficial nor harmful in itself: its effect is fixed by the projection of the displacement it removes onto the discriminative direction. On sleep-EDF the composition term costs 0.0715 of balanced accuracy and the within-class term is worth +0.0202, so removing the whole displacement costs 0.1023 in all 32 subjects, while the same operation gains 0.214 on SEED.

The mechanism is supported by an exact decomposition of the estimator that is used and by a controlled factorial that separates composition from drift. The physiological origin of the composition change is not identified, the available history in sleep-EDF is too short to test a sign change, and no label-free rule tested here predicts the sign. The grasp-and-lift negative transfer was an artefact of class-sorted storage and disappears in acquisition order: a warning about evaluation protocol, not a brain-signal mechanism.

CRediT authorship contribution statement

Zihao Gong: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing – original draft, Visualization. **Jiran Li:** Conceptualization, Methodology, Validation, Writing – review & editing. **Ke Huo:** Software, Data curation, Investigation. **Yubin Han:** Formal analysis, Validation, Visualization. **Guangyu Shi:** Data curation, Resources, Validation. **Yongqiang Xie:** Conceptualization, Supervision, Writing – review & editing, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The analysis code and result files are public at https://github.com/gzhgzy1234567890-lang/EEG-BSPC at commit ef8c890: the per-subject and per-cell outputs of the Lee2019-MI analysis, the grasp-and-lift cell table and acquisition-order rebuild, the sleep-EDF cells of all three runs, and the analysis scripts. Raw recordings are public and are not redistributed; the external cohort was read from a configurable data directory (60.4 GB). Commit ef8c890 holds the anchored decomposition, the dense strength grid and every result file quoted here; later commits change only the manuscript text.

Acknowledgements

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of AI-assisted preparation

OpenAI ChatGPT (GPT-6) was used for language editing, reference formatting, drafting of descriptive passages, and for drafting and running the Python scripts that produced the archived result files, as described under Implementation and AI assistance. All five figures are data plots produced by those scripts with Matplotlib; the layout of Fig. 1(a) and the schematic in Fig. 2(a) were drafted with AI assistance and re-rendered from the archived numbers. No figure or graphical abstract panel was produced by an image-generation model. The scientific decisions were the authors, who reviewed the code, inspected every run and verified each reported

number against the archived outputs. Those checks confirm that the archive reproduces the reported values; they are not an independent re-run of the EEG pipeline.

References

[1] M. Krauledat, M. Tangermann, B. Blankertz, K.-R. Müller, Towards zero training for brain-computer interfacing, PLoS ONE 3 (2008) e2967.

[2] F. Lotte, Signal processing approaches to minimize or suppress calibration time in oscillatory activity-based brain-computer interfaces, Proc. IEEE 103 (2015) 871–890.

[3] C. Vidaurre, C. Sannelli, K.-R. Müller, B. Blankertz, Machine-learning-based coadaptive calibration for brain-computer interfaces, Neural Comput. 23 (2011) 791–816.

[4] R. Apicella, F. Isgrò, A. Pollastro, R. Prevete, On the effects of data normalization on EEG-based BCI classification, Eng. Appl. Artif. Intell. 123 (2023) 106205.

[5] M. Congedo, A. Barachant, R. Bhatia, Riemannian geometry for EEG-based brain-computer interfaces: a primer and a review, Brain-Comput. Interfaces 4 (2017) 155–174.

[6] H. He, D. Wu, Transfer learning for brain-computer interfaces: a Euclidean space data alignment approach, IEEE Trans. Biomed. Eng. 67 (2020) 399–410.

[7] F. Yger, M. Berar, F. Lotte, Riemannian approaches in brain-computer interfaces: a review, IEEE Trans. Neural Syst. Rehabil. Eng. 25 (2017) 1753–1762.

[8] P. Zanini, M. Congedo, C. Jutten, S. Said, Y. Berthoumieu, Transfer learning: a Riemannian geometry framework with applications to brain-computer interfaces, IEEE Trans. Biomed. Eng. 65 (2018) 1107–1116.

[9] W. Samek, K.-R. Müller, M. Kawanabe, C. Vidaurre, Brain-computer interfacing in discriminative and stationary subspaces, in: Proc. IEEE EMBS, 2012, pp. 2873–2876.

[10] B. Blankertz, M. Kawanabe, R. Tomioka, F. U. Hohlefeld, V. V. Nikulin, K.-R. Müller, Invariant common spatial patterns: alleviating nonstationarities in brain-computer interfacing, in: Adv. Neural Inf. Process. Syst. 20, 2008, pp. 113–120.

[11] O. Ledoit, M. Wolf, A well-conditioned estimator for large-dimensional covariance matrices, J. Multivar. Anal. 88 (2004) 365–411.

[12] M. Saerens, P. Latinne, C. Decaestecker, Adjusting the outputs of a classifier to new a priori probabilities: a simple procedure, Neural Comput. 14 (2002) 21–41.

[13] M. Tangermann, K.-R. Müller, A. Aertsen, et al., Review of the BCI competition IV, Front. Neurosci. 6 (2012) 55.

[14] W.-L. Zheng, B.-L. Lu, Investigating critical frequency bands and channels for EEG-based emotion recognition with deep neural networks, IEEE Trans. Auton. Ment. Dev. 7 (2015) 162–175.

[15] R.-N. Duan, J.-Y. Zhu, B.-L. Lu, Differential entropy feature for EEG-based emotion classification, in: Proc. 6th Int. IEEE/EMBS Conf. Neural Eng., 2013, pp. 81–84.

[16] M. Congedo, A. Barachant, A. Andreev, A Riemannian Gauss–Laplace distribution on the manifold of symmetric positive definite matrices, with application to anomaly detection and clustering, arXiv:2306.10543 (2023).

[17] F. Lotte, L. Bougrain, A. Cichocki, M. Clerc, M. Congedo, A. Rakotomamonjy, F. Yger, A review of classification algorithms for EEG-based brain-computer interfaces: a 10 year update, J. Neural Eng. 15 (2018) 031005.

[18] MOABB developers, Cross-session motor imagery example (CSP+LDA and Riemannian geometry + logistic regression, leave-one-session-out), NeuroTechX/moabb, 2024. https://github.com/NeuroTechX/moabb

[19] W.-L. Zheng, W. Liu, Y. Lu, B.-L. Lu, A. Cichocki, EmotionMeter: a multimodal framework for recognizing human emotions, IEEE Trans. Cybern. 49 (2019) 1110–1122.

[20] M. Rahimipour, M. M. Van Hulle, Simple geometric recentering rivals deep sequence models for cross-session EEG motor-imagery decoding, bioRxiv (2026) 2026.07.07.736991, version 1, posted 11 July 2026, accessed 28 October 2026. https://doi.org/10.64898/2026.07.07.736991 (preprint).

[21] A. Bifet, R. Gavaldà, Learning from time-changing data with adaptive windowing, in: Proceedings of the 2007 SIAM International Conference on Data Mining, 2007, pp. 443–448.

[22] R.E. Kalman, A new approach to linear filtering and prediction problems, Journal of Basic Engineering 82 (1960) 35–45.

[23] S. Li, M. Kawanabe, R.J. Kobler, SPDIM: source-free unsupervised conditional and label shift adaptation in EEG, in: Proceedings of the 13th International Conference on Learning Representations (ICLR), Singapore, 2025.

[24] Z.C. Lipton, Y.-X. Wang, A. Smola, Detecting and correcting for label shift with black box predictors, in: Proceedings of the 35th International Conference on Machine Learning, 2018, pp. 3122-3130.

[25] H. Lee, et al., NeuroAdapt-Bench: a benchmark for test-time adaptation of EEG foundation models, in: Proceedings of Machine Learning for Healthcare, PMLR 340, 2026.

[26] C. Brunner, R. Leeb, G.R. Mueller-Putz, A. Schloegl, G. Pfurtscheller, BCI Competition 2008 - Graz data set A, Graz University of Technology, 2008.